Trio64V2/DX & GX Graphics/Video Accelerators

Preliminary

April 1997

S3 Incorporated
P.O. Box 58058
Santa Clara, CA 95052-8058

NOTATIONAL CONVENTIONS

The following notational conventions are used in this data book:

Signal names are shown in all uppercase letters. For example, XD.

A bar over a signal name indicates an active low signal. For example, $\overline{OE}$.

n-m indicates a bit field from bit n to bit m. For example, 7-0 specifies bits 7 through 0, inclusive.

n:m indicates a signal (pin) range from n to m. For example D[7:0] specifies data lines 7 through 0, inclusive

Use of a trailing letter H indicates a hexadecimal number. For example, 7AH is a hexadecimal number.

Use of a trailing letter b indicates a binary number. For example, 010b is a binary number.

When numerical modifiers such as K or M are used, they refer to binary rather than decimal form. Thus, for example, 1 KByte would be equivalent to 1024, not 1,000 bytes.

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Additional information may be obtained from:

S3 Incorporated, Literature Department P.O. Box 58058, Santa Clara, CA 95052-8058.

Telephone: 408-588-8000, Fax: 408-980-5444

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Section 1: Introduction

High-Performance Integrated Graphics/Video Accelerators

- High-performance 64-bit graphics engine
- Single cycle CPU writes to video memory
- Integrated 24-bit RAMDAC and programmable dual-clock synthesizer with 170 MHz output pixel rate
- Unique S3 Streams Processor for hardware-assisted video playback
- S3 Scenic Highway for direct interface to live video and MPEG-1 peripherals
- Trio64V2/DX™ is a DRAM-based pin compatible upgrade to the Trio64V+™
- Trio64V2/GX™ is a higher performance SGRAM/DRAM-based chip with a pinout optimized for synchronous operation

S3 Streams Processor Features

- On-the-fly stretching and blending of primary RGB stream and RGB or YUV (video) secondary stream
- Horizontal and vertical interpolation for smooth upscaling
- Improved secondary stream decimation
- Each stream can have different color depths
- YUV data is color space converted
- Contrast, saturation and brightness controls

Advanced Playback Capabilities

- High-quality hardware-assisted video playback (up to 1024x768x16 bits/pixel)
- Support for Indeo™ , Cinepak™ , and software-accelerated MPEG-1 video playback

Game and Presentation Effects.

- Hardware double-buffering support for high-quality tear-free playback
- 2-D scrolling and sprite plane support
- Color and chroma keying for overlaying of graphics onto video and video onto graphics
- Arithmetic blending of two pixel streams for fade-in/fade-out transition effects

S3 Scenic Highway Interface

- Industry standard video digitizers
- MPEG-1 audio/video decoders

S3 Trio64V+ Compatibility (DX)

- Pin compatible with fast page/EDO DRAMs
- BIOS and driver backwards compatibility
- Same 208-pin PQFP package

High Non-Interlaced Screen Resolution Support

- 1280x1024x256 colors at 85 Hz refresh
- 1024x768x64K colors at 85 Hz refresh
- 800x600x16.7M colors at 85 Hz refresh

High-Performance Memory Interface

- 64-bit interface
- Supports 1-, 2-, or 4-MByte frame buffer (4 MBytes only with fast page/EDO DRAMs)
- Supports fast page mode DRAM (66 MHz), 1-cycle EDO DRAM (60 MHz - 480 MB/s bandwidth), 256Kx32 SGRAM (66 MHz - 528 MB/s bandwidth)

Supports CPUs with Big or Little Endian Byte Ordering

Glueless PCI 2.1 Local Bus Support

Multimedia Support Hooks

- 8-bit bidirectional feature connector
- S3 Scenic Highway
- Serial communications bus (I^2C)

Full Software Support

- Drivers for Windows® 3.11, Windows NT™ , Windows 95™ , OS/2® 2.1 and 3.0 (Warp™), SCO® UNIX®

Green PC/Monitor Plug and Play Support

- Full hardware and BIOS support for VESA® Display Power Management Signaling (DPMS) monitor power savings modes
- DDC monitor communications support

Extensive Static/Dynamic Power Management

1.1 OVERVIEW

The S3® Trio64V2/DX and Trio64V2/GX graphics/video accelerators (hereinafter referred to as the Trio64V2/DX and Trio64V2/GX respectively), combine high-performance graphics and high-quality video acceleration features with the capability to directly interface to live video and MPEG-1 peripherals. They incorporate an enhanced version of the 64-bit graphics accelerator core and high-performance 170 MHz true-color RAMDAC that are found in S3's existing Trio64V+ accelerator. Note that the term Trio64V2™ graphics/video accelerator (or Trio64V2) is used when the discussion applies equally to the Trio64V2/DX and Trio64V2/GX.

All display applications that require high-quality video playback (from a CD-ROM or hard drive), or live video input capability, can take advantage of the new features. The Trio64V2 accelerates/enhances software MPEG-1/Indeo/Cinepak video playback by providing arbitrary scaling with high-quality linear interpolation in both the horizontal and vertical dimensions and color space conversion (RGB to YUV). By performing these tasks in hardware and relieving the CPU of a substantial overhead, the Trio64V2 offers high-quality video playback with window sizes of up to 1024x768x16 bits/pixel at high frame rates. The Trio64V2 has an S3 Scenic Highway™ interface that provides a direct interface to MPEG-1 audio/video decoder devices and live video digitizers to attain full-motion video.

1.2 S3 Streams Processor

The S3 Streams Processor™ allows the mixing of three separate display streams. The primary stream can be RGB data of any color depth. The secondary stream can be RGB or YUV (video) data of any color depth. YUV data is color space converted to RGB. The third stream, the hardware cursor, overlays the other two streams. Horizontal and vertical interpolation allow smooth upscaling of the secondary stream.

Arithmetic blending of a primary graphics stream and secondary graphics/video enables dramatic transition effects for game applications. Color and chroma keying allow opaque or transparent overlays of one stream on the other. Hardware-assisted double buffering of both primary and secondary data streams is also provided to enable high-quality "tear-free" playback.

The Trio64V2 also enhances game acceleration, with support for a sprite plane where sprites are actually rendered into a sprite plane memory. Sprites can be overlaid onto the background without saving and restoring the background.

The Streams Processor is located in the pixel datapath between the display memory and the RAMDAC that drives the RGB signals to the monitor. One of the key advantages of this architecture is that it permits processing of pixel streams on the fly at display refresh rates. This eliminates the need to first write back processed (scaled or color-space-converted) data into the frame buffer before sending it to the RAMDAC. This saves memory storage and memory bandwidth.

The Streams Processor also enables simultaneous display of graphics and video of different color depths. For example, it is possible to display 24 bits/pixel-equivalent video on top of an 8-bit graphics background. This also saves memory bandwidth and storage capacity while permitting higher frame rates because of reduced bandwidth requirements.

In addition, if an opaque rectangular window of one stream is overlaid onto another background window of a second stream, it is not necessary to fetch and refresh the hidden pixels. This provides additional memory bandwidth savings.

The contrast, saturation and brightness of the video stream can be adjusted. This allows the viewer to adjust for personal preferences.

1.3 S3 Scenic Highway

The S3 Scenic Highway interface directly connects to standard MPEG-1 audio/video decoders as well as industry standard video digitizers. This provides easy implementation of MPEG-1 or digital video daughtercards that directly plug into the Scenic Highway connector or, alternately, ISA cards, where a ribbon cable is also necessary.

The Streams Processor and Scenic Highway are tightly coupled to provide optimal live video playback. The hardware automatically switches capture and display buffers without software intervention.

1.4 CHANGES FROM THE Trio64V+

The Trio64V2/DX is a pin-compatible and software-backward-compatible upgrade to the Trio64V+. It has the same DRAM options as the Trio64V+ and is designed primarily to provide improved video performance. The Trio64V2/GX has the same functionality as the Trio64V2/DX except that it provides greater memory bandwidth by using SGRAMs. The Trio64V2/GX pinout is optimized for SGRAMs and is not compatible with the DX pinout. However, the Trio64V2/GX can be used with fast page or EDO DRAMs.

The significant enhancements to the Trio64V+ are:

- Vertical filtering is added
- LPB decimation is changed to provide better handling of YUV/YCbCr data in addition to the Trio64V+ decimation scheme
- New Streams FIFO fetching scheme based on separate N values for the Primary and Secondary Streams and new FIFO fetch start timing
- Video brightness, contrast and saturation adjustment is provided
- Color keying can be based on the index or color value for 8 bpp modes
- SGRAMs are supported at 66 MHz (1 or 2 MBytes)
- Fast page mode operation is supported at 66 MHz and 1-cycle EDO operation is supported at 60 MHz.
- End of line and end of frame indicators automatically generated in LPB Pass-Through mode
- Odd/Even field detection provided for video input
- Blank pedestal RAMDAC support is added
- 1-cycle linear addressing for CPU accesses to video memory
- PCI subsystem ID and subsystem vendor ID supported
- PD output drive programmable between 4 and 8 ma

The following Trio64V+ features are not supported:

- VL-Bus
- CL-480 interface
- Trio64-compatible VAFC
- Trio64-compatible mode

1.5 ABOUT THIS DATA BOOKLET

For the most part, this data booklet describes how the Trio64V2 differs from the Trio64V+. Therefore, users will also need the Trio64V+ data book. For convenience, the Pins, Electrical Data and Mechanical Data sections of this data booklet are complete.

Section 2: Pins

2.1 PINOUT DIAGRAMS

The Trio64V2/DX and Trio64V2/GX each comes in a 208-pin PQFP package. The Trio64V2/DX provides a Trio64V+-compatible pinout for use with fast page/EDO memory and is shown in Figure 2-1. The pinout for the Trio64V2/GX is optimized for use of SGRAM memory and is shown in Figure 2-2. The Trio64V2/GX can also be used with fast page/EDO memory.

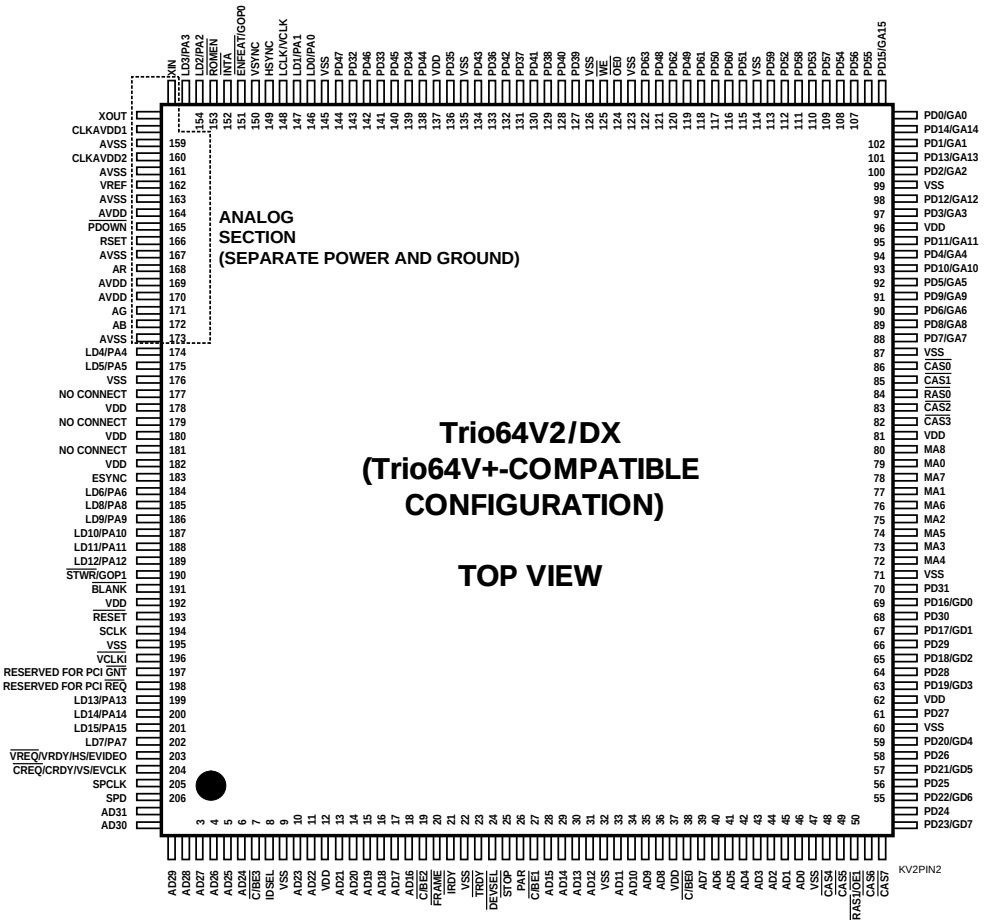
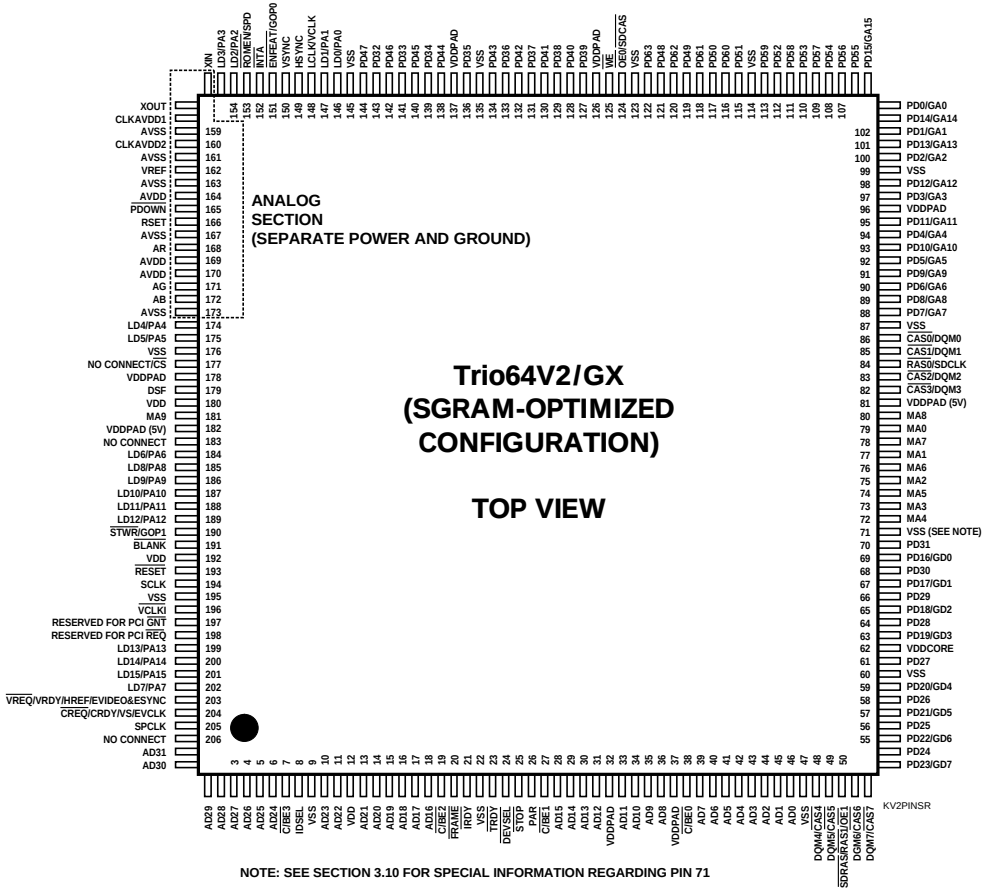


Figure 2-1. Trio64V2/DX Pinout



NOTE: SEE SECTION 3.10 FOR SPECIAL INFORMATION REGARDING PIN 71

Figure 2-2. Trio64V2/GX Pinout

2.2 PIN DESCRIPTIONS

The following table provides a brief description of each pin. The following abbreviations are used for pin types.

I - Input signal
O - Output signal
B - Bidirectional signal

Table 2-1. Pin Descriptions

Symbol	Type	Description
PCI BUS INTERFACE		
Address and Data		
AD[31:0]	B	Multiplexed Address/Data Bus. A bus transaction (cycle) consists of an address phase followed by one or more data phases.
$\overline{\text{C/BE[3:0]}}$	I	Bus Command/Data Byte Enables. These signals carry the bus command during the address phase and the byte enables during the data phase.
Bus Control		
SCLK	I	PCI System Clock.
$\overline{\text{INTA}}$	O	Interrupt Request.
$\overline{\text{IRDY}}$	I	Initiator Ready. A bus data phase is completed when both $\overline{\text{IRDY}}$ and $\overline{\text{TRDY}}$ are asserted on the same cycle.
$\overline{\text{TRDY}}$	O	Target Ready. A bus data phase is completed when both $\overline{\text{IRDY}}$ and $\overline{\text{TRDY}}$ are asserted on the same cycle.
$\overline{\text{DEVSEL}}$	O	Device Select. Trio64V2 drives this signal active when it decodes its address as the target of the current access.
IDSEL	I	Initialization Device Select. This input is the chip select for PCI configuration register reads/writes.
$\overline{\text{RESET}}$	I	System Reset. Asserting this signal forces the registers and state machines to a known state.
$\overline{\text{FRAME}}$	I	Cycle Frame. This signal is asserted by the bus master to indicate the beginning of a bus transaction. It is deasserted during the final data phase of a bus transaction.
PAR	O	Parity. Trio64V2 asserts this signal to verify even parity during reads.
$\overline{\text{STOP}}$	O	Stop. Trio64V2 asserts this signal to indicate a target disconnect or retry.

Table 2-1. Pin Descriptions (continued)

Symbol	Type	Description
CLOCK SOURCE		+
XIN	I	Reference Frequency Input. If an external crystal is used, it is connected between XOUT and this pin. A stable external frequency source can also be input via this pin. If this is done, the input voltage must be the same as AVDD. If PD11 is strapped high at power-on, this becomes the DCLK (dot clock) input, bypassing the internal oscillator. This is normally only used for test purposes.
XOUT	O	Crystal Output. If an external 14.318 MHz crystal is used, it is connected between XIN and this pin. This pin drives the crystal via an internal oscillator.
DISPLAY MEMORY INTERFACE		
Address and Data		
MA[9:0]	O	Memory Address Bus. The memory row and column addresses are multiplexed on these lines. MA9 is the SGRAM/SDRAM bank select pin for the Trio64V2/GX.
PD[63:0]	B	Memory Data Bus Lines.
DRAM Memory Control		
$\overline{\text{RAS}}$[1:0]	O	Row Address Strobe
$\overline{\text{CAS}}$[7:0]	O	Column Address Strobes.
$\overline{\text{WE}}$	O	Write Enable.
$\overline{\text{OE}}$[1:0]	O	Output Enables.
SGRAM Memory Control (Trio64V2/GX)		
SDCLK	O	SGRAM Clock.
$\overline{\text{SDRAS}}$	O	SGRAM Row Address Strobe.
$\overline{\text{SDCAS}}$	O	SDRAM Column Address Strobe.
DQM[7:0]	O	SGRAM Data Input/Output Masks.
$\overline{\text{CS}}$	O	SGRAM Chip Select.
DSF	O	SGRAM Special Function.
$\overline{\text{WE}}$	O	Write Enable.
VIDEO INTERFACE		
$\overline{\text{PDOWN}}$	I	Power Down. Asserting this signal turns off the RGB analog output from the DACs.
VREF		Voltage Reference. This pin is tied to V_{SS} through a 0.1 µF capacitor.
RSET		Reference Resistor. This pin is tied to V_{SS} through an external resistor to control the full-scale current value.
AR	O	Analog Red. Analog red output to the monitor.
AG	O	Analog Green. Analog green output to the monitor.
AB	O	Analog Blue. Analog blue signal to the monitor.

Table 2-1. Pin Descriptions (continued)

Symbol	Type	Description
$\overline{\text{ENFEAT}}$	O	Enable Feature Connector. Setting SRD_0 to 1 drives this signal low when SR1C_1-0 are 00b. This also enables all feature connector operations.
$\overline{\text{BLANK}}$	B	Video Blank.
ESYNC	I	External SYNC. When ESYNC is driven low, HSYNC, VSYNC and $\overline{\text{BLANK}}$ become inputs. When ESYNC is high, HSYNC, VSYNC and $\overline{\text{BLANK}}$ become outputs. For the Trio64V2/GX, this signal is the same as EVIDEO.
EVIDEO	I	External Video. When this input is high, PA[15:0] are outputs to the feature connector. For the Trio64V2/GX, this signal is the same as ESYNC.
EVCLK	I	External VCLK. When this input is asserted low, VCLK is an input to the internal RAMDAC. When this input is high, VCLK is output to the feature connector.
VCLK	B	Video/Pixel Clock. When EVCLK is high, this signal is an output to the feature connector. When EVCLK is low, this becomes an input used only for test purposes.
VCLKI	I	VCLK Input. The VCLKI function is enabled when LPB VAFC (16-bit) feature connector operation is enabled. Setting bit 1 of SRB to 1 causes VCLKI to be used to clock in feature connector pixel data to the internal RAMDAC.
HSYNC	B	Horizontal Sync. When ESYNC is high, this is the horizontal sync output. When ESYNC is low, this is an input from the feature connector.
VSYNC	B	Vertical Sync. When ESYNC is high, this is the vertical sync output. When ESYNC is low, this is an input from the feature connector.
PA[15:0]	B	Pixel Address Lines [15:0]. These are the data lines when feature connector operation is enabled. When EVIDEO is high, PA signals are outputs to the feature connector.

Table 2-1. Pin Descriptions (Continued)

Symbol	Type	Description
MISCELLANEOUS FUNCTIONS		
General Data, I/O and Serial Ports		
GA[15:0]	O	ROM Address Bus. These signals provide the address for BIOS ROM reads. They are multiplexed with MD signals. Programmers must ensure that the memory bus is inactive when reading the ROM.
GD[7:0]	I	ROM Data Bus. These signals carry data for BIOS ROM reads. They are multiplexed with MD signals. Programmers must ensure that the memory bus is inactive when reading the ROM.
$\overline{\text{ROMEN}}$	O	ROM Enable. This signal provides the chip output enable input for BIOS ROM reads.
GOP[1:0]	O	General Output Port Bits 1-0. If SR1C_1 is set to 1, the value of CR5C_0 is output on pin 151 (GOP0) and the value of CR5C_1 is output on pin 190 (GOP1).
$\overline{\text{STWR}}$	O	Strobe Write. If SR1C_1 is cleared to 0, this signal is asserted whenever a write is made to CR5C. It is used to enable a General Output Port latch.
SPCLK	I/O	Serial Port Clock. This is the clock for serial data transfer, either for I^2C or DDC2 monitor data communications. As an output, it is programmed via MMFF20_0. As an input, its status is read via MMFF20_2. In either case the serial port must be enabled by setting MMFF20_4 to 1.
SPD	I/O	Serial Port Data. This is the data signal for serial data transfer, either for I^2C or DDC2 monitor data communications. As an output, it is programmed via MMFF20_1. As an input, its status is read via MMFF20_3. In either case the serial port must be enabled by setting MMFF20_4 to 1.
SCENIC HIGHWAY		
Bidirectional Mode		
LD[7:0]	I/O	LPB Data. This is the Scenic Highway data bus.
LCLK	I	LPB Clock. This clock controls transactions between the Trio64V2 and Scenic Highway peripherals
$\overline{\text{VREQ}}$/VRDY	O	Video Request/Ready.
$\overline{\text{CREQ}}$/CRDY	I	Decoder Request/Ready.
Video 8 and 16 Modes		
LD[15:0]	I	LPB Data Bus [15:0]. Video data input. LD[7:0] are used in Video 8 mode.
HREF	I	HSYNC input signaling the transition from one line to the next.
VS	I	VSYNC input signaling the transition from one frame to the next.

Table 2-1. Pin Descriptions (Continued)

Symbol	Type	Description
POWER AND GROUND		
VDD	I	Digital power (5V)
VDDPAD	I	Pad digital power (5V for Trio64V2/DX and Trio64V2/GX using EDO meomory. 3.3V for Trio64V2/GX using synchronous memory.)
AVDD	I	Analog power supply (RAMDAC).
CLKAVDD[1:2]	I	Analog power supply (clock synthesizer, 1 = DCLK, 2 = MCLK)
VSS	I	Digital ground
AVSS	I	Analog ground

2.3 PIN LISTS

Table 2-2 lists all pins alphabetically. Table 2-3 lists all pins in numerical order.

Table 2-2. Alphabetical Pin Listing

Name	PIN(S)
AB	172
AD[31:16]	207, 208, 1, 2, 3, 4, 5, 6, 10, 11, 13, 14, 15,16, 17, 18
AD[15:0]	28, 29, 30, 31, 33, 34, 35, 36, 39, 40, 41, 42, 43, 44, 45, 46
AG	171
AR	168
AVDD	164, 169, 170
AVSS	159, 161, 163, 167, 173
$\overline{\text{BLANK}}$	191
$\overline{\text{CAS}}$[7:0]	52, 51, 49, 48, 82, 83, 85, 86
$\overline{\text{C/BE}}$[3:0]	7, 19, 27, 38
CLKAVDD[1:2]	158, 160
$\overline{\text{CREQ}}$/CRDY	204
CS	177 (GX)
$\overline{\text{DEVSEL}}$	24
DQM[7:0]	52, 51, 49, 48, 82, 83, 85, 86 (GX)
DSF	179 (GX)
$\overline{\text{ENFEAT}}$	151
ESYNC	203
EVCLK	204
EVIDEO	203
$\overline{\text{FRAME}}$	20
GA[15:0]	105, 103, 101, 98, 95, 93, 91, 89, 88, 90, 92, 94, 97, 100, 102, 104
GD[7:0]	53, 55, 57, 59, 63, 65, 67, 69
GOP[1:0]	190, 151
HREF	203
HSYNC	149
IDSEL	8
$\overline{\text{INTA}}$	152
$\overline{\text{IRDY}}$	21
LCLK	148
LD[15:0]	201, 200, 199, 189, 188, 187, 186, 185, 202, 184, 175, 174, 155, 154, 147, 146
MA[9:0]	181, 80, 78, 76, 74, 72, 73, 75, 77, 79 (MA9 only on GX)
NO CONNECT	183, 206
$\overline{\text{OE}}$[1:0]	50, 124
PA[15:0]	201, 200, 199, 189, 188, 187, 186, 185, 202, 184, 175, 174, 155, 154, 147, 146
PAR	26

Table 2-2. Alphabetical Pin Listing (Continued)

Name	PIN(S)
PD[63:48]	122, 120, 118, 116, 113, 111, 109, 107, 106, 108, 110, 112, 115, 117, 119, 121
PD[47:32]	144, 142, 140, 138, 134, 132, 130, 128, 127, 129, 131, 133, 136, 139, 141, 143
PD[31:16]	70, 68, 66, 64, 61, 58, 56, 54, 53, 55, 57, 59, 63, 65, 67, 69
PD[15:0]	105, 103, 101, 98, 95, 93, 91, 89, 88, 90, 92, 94, 97, 100, 102, 104
$\overline{\text{PDOWN}}$	165
$\overline{\text{RAS}}$[1:0]	50, 84
RESERVED	197, 198
$\overline{\text{RESET}}$	193
$\overline{\text{ROMEN}}$	153
RSET	166
SCLK	194
$\overline{\text{SDCAS}}$	124 (GX)
SDCLK	84 (GX)
$\overline{\text{SDRAS}}$	50 (GX)
SPCLK	205
SPD	206 (DX), 153 (GX)
$\overline{\text{STOP}}$	25
$\overline{\text{STWR}}$	190
$\overline{\text{TRDY}}$	23
VCLK	148
VCLKI	196
VDD	12, 62, 180, 192
VDDPAD	32 (GX), 37, 96, 126 (GX), 137, 178
VDDPAD (5V)	81, 182
$\overline{\text{VREQ}}$/VRDY	203
VREF	162
VS	204
VSS	9, 22, 32 (DX), 47, 60, 71, 87, 99, 114, 123, 126 (DX), 135,
	145, 176, 195
VSYNC	150
$\overline{\text{WE}}$	125
XIN	156
XOUT	157

Table 2-3. Numerical Pin Listing

Number	Pin	Number	Pin
1	AD29	43	AD3
2	AD28	44	AD2
3	AD27	45	AD1
4	AD26	46	AD0
5	AD25	**47**	**VSS**
6	AD24	48	CAS4/DQM4
7	C/BE3	49	CAS5/DQM5
8	IDSEL	50	RAS1/OE1/SDRAS
9	**VSS**	51	CAS6/DQM6
10	AD23	52	CAS7/DQM7
11	AD22	53	PD23/GD7
12	**VDD**	54	PD24
13	AD21	55	PD22/GD6
14	AD20	56	PD25
15	AD19	57	PD21/GD5
16	AD18	58	PD26
17	AD17	59	PD20/GD4
18	AD16	**60**	**VSS**
19	C/BE2	61	PD27
20	FRAME	**62**	**VDD**
21	IRDY	63	PD19/GD3
22	**VSS**	64	PD28
23	TRDY	65	PD18/GD2
24	DEVSEL	66	PD29
25	STOP	67	PD17/GD1
26	PAR	68	PD30
27	C/BE1	69	PD16/GD0
28	AD15	70	PD31
29	AD14	**71**	**VSS**
30	AD13	72	MA4
31	AD12	73	MA3
32	**VSS//VDDPAD (Note 1)**	74	MA5
33	AD11	75	MA2
34	AD10	76	MA6
35	AD9	77	MA1
36	AD8	78	MA7
37	**VDDPAD**	79	MA0
38	C/BE0	80	MA8
39	AD7	**81**	**VDDPAD (5V)**
40	AD6	82	CAS3/DQM3
41	AD5	83	CAS2/DQM2
42	AD4	84	RAS0/SDCLK

Table 2-3. Numerical Pin Listing (Continued)

Number	Pin	Number	Pin
85	CAS1/DQM1	127	PD39
86	CAS0/DQM0	128	PD40
87	**VSS**	129	PD38
88	PD7/GA7	130	PD41
89	PD8/GA8	131	PD37
90	PD6/GA6	132	PD42
91	PD9/GA9	133	PD36
92	PD5/GA5	134	PD43
93	PD10/GA10	**135**	**VSS**
94	PD4/GA4	136	PD35
95	PD11/GA11	**137**	**VDDPAD**
96	**VDDPAD**	138	PD44
97	PD3/GA3	139	PD34
98	PD12/GA12	140	PD45
99	**VSS**	141	PD33
100	PD2/GA2	142	PD46
101	PD13/GA13	143	PD32
102	PD1/GA1	144	PD47
103	PD14/GA14	**145**	**VSS**
104	PD0/GA0	146	LD0/PA0
105	PD15/GA15	147	LD1/PA1
106	PD55	148	LCLK/VCLK
107	PD56	149	HSYNC
108	PD54	150	VSYNC
109	PD57	151	ENFEAT/GOP0
110	PD53	152	INTA
111	PD58	153	ROMEN//ROMEN/SPD (Note 1)
112	PD52	154	LD2/PA2
113	PD59	155	LD3/PA3
114	**VSS**	156	XIN
115	PD51	157	XOUT
116	PD60	**158**	**CLKAVDD1**
117	PD50	**159**	**AVSS**
118	PD61	**160**	**CLKAVDD2**
119	PD49	**161**	**AVSS**
120	PD62	162	VREF
121	PD48	**163**	**AVSS**
122	PD63	**164**	**AVDD**
123	**VSS**	165	PDOWN
124	OE0/SDCAS	166	RSET
125	WE	**167**	**AVSS**
126	**VSS//VDDPAD (Note 1)**	168	AR

Table 2-3. Numerical Pin Listing (Continued)

Number	Pin	Number	Pin
169	**AVDD**	189	LD12/PA12
170	**AVDD**	190	$\overline{\text{STWR}}$/GOP1
171	AG	191	$\overline{\text{BLANK}}$
172	AB	**192**	**VDD**
173	**AVSS**	193	$\overline{\text{RESET}}$
174	LD4/PA4	194	SCLK
175	LD5/PA5	**195**	**VSS**
176	**VSS**	196	VCLKI
177	NO CONNECT//$\overline{\text{CS}}$ (Note 1)	197	Reserved for PCI $\overline{\text{GNT}}$
178	**VDDPAD**	198	Reserved for PCI $\overline{\text{REQ}}$
179	NO CONNECT//DSF (Note 1)	199	LD13/PA13
180	**VDD**	200	LD14/PA14
181	NO CONNECT//MA9 (Note 1)	201	LD15/PA15
182	**VDD (5V for GX)**	202	LD7/PA7
183	ESYNC//NO CONNECT (Note 1)	203	$\overline{\text{VREQ}}$/VRDY/HREF/EVIDEO&ESYNC
184	LD6/PA6	204	$\overline{\text{CREQ}}$/CRDY/VS/EVCLK
185	LD8/PA8	205	SPCLK
186	LD9/PA9	206	SPD//NO CONNECT (Note 1)
187	LD10/PA10	207	AD31
188	LD11/PA11	208	AD30

Note

1. Listings with two entries separated by // are for the Trio64V2/DX and Trio64V2/GX configurations respectively.

Section 3: Functional Changes from the Trio64V+

3.1 MEMORY SUPPORT

The Trio64V2/GX adds support for SGRAMs operating at 66 MHz. Fast page or EDO DRAMs are still supported as an alternative and can be operated at a higher MCLK speed than allowed by the Trio64V+. This is true for both the Trio64V2/DX and Trio64V2/GX.

3.1.1 SGRAM Support (Trio64V2/GX Only)

SGRAM support is enabled via power-on strapping bits CR36_32 = 01b. SGRAMs require that their operating mode be programmed after reset. The following register bits are added to support this requirement.

CR88

Bits 4-3 SGRAM $\overline{CAS}$ Latency
00 = Reserved
01 = Reserved (for 1 MCLK)
10 = 3 MCLKs
11 = 2 MCLKs

Bit 5 SGRAM Set Mode Register

Bit 6 Disable RAM Refresh

During each power-on cycle, the BIOS must first disable RAM refresh by setting CR88_6 to 1. After the $\overline{CAS}$ latency is programmed, CR88_5 is programmed to 1 to generate a mode programming cycle. This bit is automatically cleared to 0 after the programming cycle. DRAM refresh is then enabled by clearing CR88_6 to 0.

Special SGRAM features such as block write and write-per-bit are not supported. 4-MByte SGRAM/SDRAM configuration are not supported (only 1 or 2 MBytes). A 2-MByte SGRAM configuration is shown in Figure 3-1.

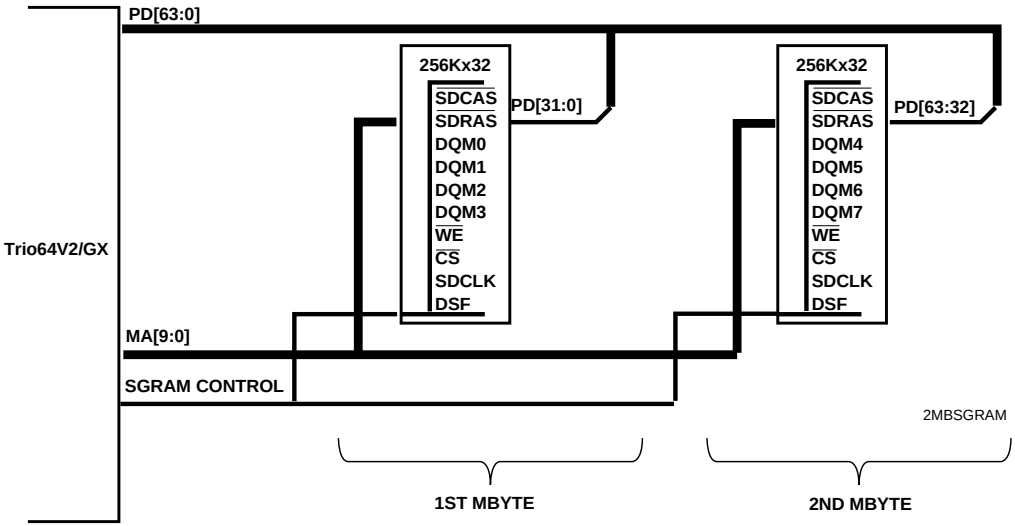


Figure 3-1. 2-MByte SGRAM Configuration

3.1.2 DRAM Speed

The maximum MCLK rate is 66 MHz for fast page operation and 60 MHz for 1-cycle EDO operation.

3.2 VIDEO DISPLAY ENHANCEMENTS

The Trio64V2 provides a number of enhancements for displaying video in a secondary window.

3.2.1 Decimation

In the horizontal decimation scheme presently implemented in the Trio64V+, each bit in the horizontal decimation mask register corresponds to a byte in Video 8 Mode, 2 bytes in Video 16 mode and 4 bytes in pass-through mode. Any type of data could be decimated. This type of decimation is supported by the Trio64V2 when MMFF00_7 = 0. If MMFF00_7 is set to 1, an alternate decimation scheme is used that is better suited for YUV/YCbCr data.

The new decimation scheme works as follows:

1. Only YUV 4:2:2 data should be decimated.
2. All LPB modes (Video 8, Video 16, pass-through) use the same decimation scheme.
3. The LPB Horizontal Decimation Control register (MMFF2C) is programmed, with each bit corresponding to the Y of a YU or YV pair. A 0 specifies that the corresponding Y (luma) be kept and a 1 specifies that the corresponding Y be dropped.
4. The number of 0's programmed in MMFF2C must be 0 or a multiple of 4.

5. Kept Y's are paired sequentially, with each Y pair being assigned the UV pair associated with the first Y in the pair.

The starting conditions for decimation are the same as for the Trio64V+.

3.2.2 Vertical Filtering

Vertical filtering significantly improves the appearance of stretched video windows. Use of vertical filtering of video data requires that CR23 and CR26 be programmed per the CR26 register description in Section 4 and that the following new register bit be set.

MM81E8_15
0 = Line replication used
1 = Vertical filtering (interpolation) enabled

In general, there are no restrictions on the use of vertical filtering. However, in the following cases:

1 MByte, 40 MHz 1-cycle EDO @ 75 Hz
1 MByte, 66 MHz fast page @ 60 Hz

vertical filtering should not be enabled unless the horizontal stretch factor is 2 or greater.

With vertical filtering enabled, memory bandwidth savings mode can be enabled by setting MM81E8_14 to 1. This reduces line fetches. In addition, the UV/CbCr color components can be dropped by setting CR87_0 to 1. This is effective when both vertical filtering is enabled (MM81E8_15 = 1) and memory bandwidth savings mode is enabled (MM81E8_14 = 1). When vertical filtering is off, bandwidth savings can be achieved by setting MM81E8_13 = 1.

3.2.3 Automatic Line/Frame Indicators

In LPB pass-through mode, there may be no indication of end of line (HSYNC) and end of frame (VSYNC). The Trio64V2 provides a method of automatically generating these indicators via the following bit.

MMFF00_14 - Auto Sync Generation
0 = No auto sync generation
1 = HSYNC and VSYNC automatically generated based on the horizontal count programmed in MMFF28_11-0 and the vertical count programmed in MMFF28_24-16. When the horizontal count value is reached, the stride (MMFF34_11-0) is added to the address to start the next line.

3.2.4 Odd/Even Frame Detection

The Trio64V2 provides automatic detection of odd/even frame when operating in the LPB Video 8 or Video 16 mode. This is enabled via the following bit:

MMFF00_30 - Odd/Even Field Detection Enable
0 = Odd/Even field detection disabled
1 = Odd/Even field detection enabled

The detection technique is based on the protocol specified by the SAA7110, as follows:

Each time the vertical sync input (VS) transitions from high to low, the state of the horizontal sync input (HREF) is latched. A logic 0 for HREF indicates the odd/top field is being sent. A logic 1 indicates the even/bottom field is being sent. This is shown in Figure 3-2.

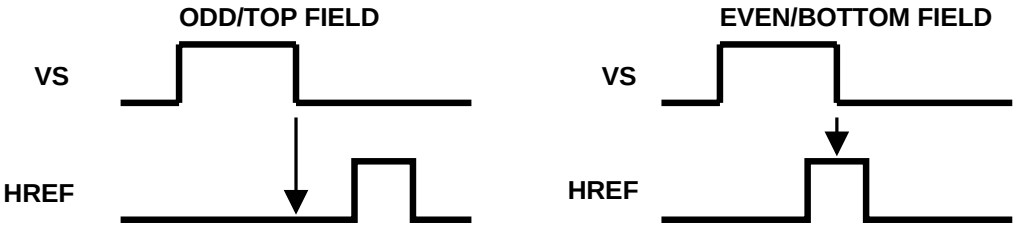


Figure 3-2. Odd/Even Field Detection

This value is EXCLUSIVE-ORed with the value of MMFF00_29 to allow polarity inversion, and the result (status) is latched in MMFF00_28.

For video input devices that do not provide sync signals in conformance with the SAA7110 protocol, alternate signals must be provided that meet the timing requirements given the AC timings in Section 5.

3.2.5 FIFO Fetching Enhancements

Several enhancements have been added to optimize the use of memory bandwidth for display FIFO fetching when a secondary video window is displayed.

In full Streams Processor mode (CR67_3-2 = 11b), display FIFO fetching automatically begins as soon as possible after the start of blanking. This starting position can be delayed by number of character clocks programmed in CR85_2-0. In other Streams modes, FIFO fetch start is controlled as for the Trio64V+ via CR3B.

A separate register (CR72_7-0) is provided for a Secondary Stream N value. This is used to ensure that adequate secondary stream pixel data is fetched each line to prevent display problems.

L parameters, which control the length of FIFO fetches, have been added for both the primary and secondary stream FIFOs. See the descriptions of CR90, CR91, CR92 and CR93. Use of these registers eliminates unnecessary FIFO fetches.

3.2.6 Color Keying on Index

The following register bit adds flexibility to the color keying capability for VGA and 8-bit enhanced modes.

MM8184_29
0 = Color keying based on color value for VGA and 8-bit enhanced modes
1 = Color keying based on comparison of color index value with the value specified in MM8184_7-0

3.3 COLOR ADJUSTMENTS

The following new register controls color adjustments. Color adjustment should be enabled only for YUV/YCbCr secondary stream formats. They must be disabled (MM819C_15 = 0 and MM819C_31 = 0) for secondary stream RGB formats.

MM819C - Color Adjustment Register

Bits 7-0 Brightness Control

Value = SBBBBBBB

where S is the sign bit (1 = negative) and BBBBBBB is the brightness adjustment factor. The larger the number, the greater the brightness. The programmed value is multiplied by 2 by the hardware. The value is in 2's complement format.

Bits 12-8 Contrast Control

Value = C.CCCC

Where C.CCCC is the contrast adjustment, which can vary from 0 to 1.9375. The value is in 2's complement format.

Bit 15 Brightness and Contrast Enable
0 = Brightness and contrast control disabled
1 = Brightness and contrast control enabled

Bit 20-16 Hue and Saturation Factor 1

Value = SF.FFF

where S is the sign bit (1 = negative) and the F.FFF is the factor [SAT * cos A]. SAT is the saturation, which can vary from -2 to 1.875 and A is the hue angle, the cosine of which varies from -1 to +1. The value is in 2's complement format.

Bits 28-24 Hue and Saturation Factor 2

Value = SF.FFF

where S is the sign bit (1 = negative) and the F.FFF is the factor [SAT * sin A]. SAT is the saturation, which can vary from -2 to 1.875 and A is the hue angle, the sine of which varies from -1 to +1. The value is in 2's complement format.

Bit 31 Hue and Saturation Control Enable
0 = Hue and saturation control disabled
1 = Hue and saturation control enabled

3.4 BLANK PEDESTAL

$\overline{\text{BLANK}}$ pedestal support is enabled by setting SR1A_5 to 1. This white level output remains at 17.6 mA (typical) above the black level output. However, the black level output is raised to 1.45 mA (typical) over the $\overline{\text{BLANK}}$ output level of 0 mA. RAMDAC output using the $\overline{\text{BLANK}}$ pedestal is recommended for some monitors.

3.5 1-CYCLE LINEAR ADDRESSING

Setting CR87_1 to 1 enables 1-cycle linear addressing of video memory. The memory must be capable of supporting this mode of operation. The default (CR87_1 = 0) is 2-cycle operation if SRA_7 and SR15_7 are set to 1.

3.6 DCLK PLL FREQUENCY

The maximum DCLK PLL frequency has been increased to 340 MHz. The DCLK PLL R divider field has been increased to 3 bits by adding SR12_7. When SR12_7-5 = 100b, the frequency divider is 16.

3.7 PD DRIVE CURRENT

The drive current of PD[63:0] is programmable via CR80_4. The default value of 0 selects 8 ma drive current; setting this bit to 1 selects 4 ma drive current.

3.8 PCI SUBSYSTEM ID

The Subsystem ID and Subsystem Vendor ID are located in a 32-bit read only register at PCI Configuration Space Index 2C. These registers reflect the content of 4 new read/write CR registers as follows:

Register	CR Space	PCI Configuration Space
Subsystem Vendor ID Low Byte	CR81	Index 2CH
Subsystem Vendor ID High Byte	CR82	Index 2DH
Subsystem ID Low Byte	CR83	Index 2EH
Subsystem ID High Byte	CR84	Index 2FH

These registers allow identification of particular vendors using the same graphics chip. The following design allows the subsystem identification to be handled by software (no hardwiring).

For add-in card cases, the system BIOS will typically attempt to read the subsystem ID information before the video BIOS is run. When this occurs, Trio64V2 will check a read/write Subsystem ID Source bit (CR37_2), which defaults to 1 on power-up. This bit is defined as follows:

CR37_2 - Subsystem ID Source

0 = Read subsystem ID information from chip registers
1 = Read subsystem ID information from BIOS

On power-up, with this bit set to 1, the following occurs:

1. The PCI bus is claimed and held by the Trio64V2 when an attempt to read the subsystem ID occurs.
2. Video ROM BIOS locations C0040-C0043 are read, depending on which $\overline{BE[0:3]}$ lines are enabled. For example, if only $\overline{BE0}$ is enabled, then the byte at C0040 will be read. The information in C0040-C0043 is editable using S3's Video BIOS edit utility, and thus can be tailored by each vendor.
3. The PCI bus is released.
4. (Some time later) The video BIOS is run. This copies the information in C0040-C0043 to CR81-CR83 (and the mirrored PCI2C-PCI2F). It also resets CR37_2 to 0. Until this step is taken, all subsystem register accesses will generate the sequence listed in steps 1-3 above.

From this point on until the next hard re-boot, all subsystem register accesses will be directed to the PCI subsystem registers. A soft reset (e.g., CTL-ALT-DEL) does not change the state of CR37_2 and thus does not affect the subsystem ID read process.

Motherboard designs incorporate the video BIOS and system BIOS in the same ROM, so care must be taken that the Trio64V2 does not generate a video ROM access cycle. To accomplish this, PD10 is pulled low at reset to latch a 0 in CR37_2. The system BIOS must then load the subsystem ID information in the Trio64V2 before any ID scanning takes place. To do this, it must turn on the Trio64V2, enable I/O accesses in the PCI configuration space, unlock the CR registers, program the subsystem ID information in the registers described above, then turn off the Trio64V2.

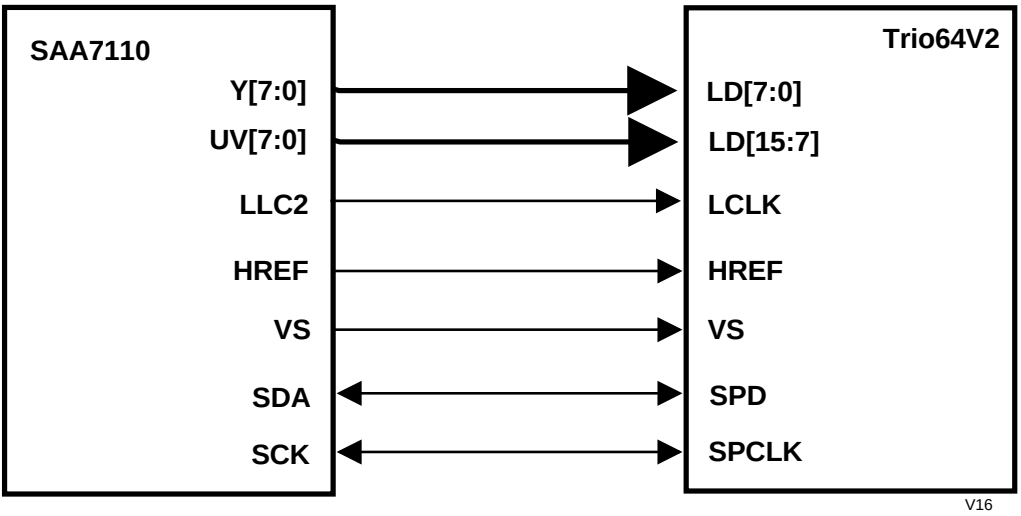


Figure 3-3. Video 16 Interface

3.9 VIDEO 16 LPB INTERFACE

The Video 16 interface is unchanged from the Trio64V+ except that HREF should be used instead of HS. However, the YUV connections are not explicitly stated in the Trio64V+ data book and are shown in Figure 3-3.

3.10 FUTURE COMPATIBILITY

For pin-compatibility with future S3 chips, special consideration must be taken with the board design regarding pin 71. Contact your S3 technical representative for details.

3.11 NAND TREE TESTING

NAND-tree scanning is enabled when PD9 is pulled low at reset. All pins in the NAND-tree list become inputs. SPCLK (pin 205) is the output pin for the scan and $\overline{\text{RESET}}$ cannot be tested and must not be connected. MA[9:0] are output-only pins and are not tested. In addition, VSYNC and the analog pins are not included in the the scan.

The tester drives high logic to the pins in the NAND tree. It then sequentially drives and holds each input low and records the state of the SPCLK pin. Testing follows the order shown in Table 3-1, starting with pin 206 and moving down each column until pin 204 is reached. If all pin connections are good, the output will be a perfect square wave. If this pattern is broken, the location of the incorrect signal level in the sequence identifies the bad connection.

Table 3-1. NAND Tree Test Order

#	Pin	#	Pin	#	Pin
206	SPD	054	PD24	121	PD48
207	AD31	055	PD22/GD6	122	PD63
208	AD30	056	PD25	127	PD39
001	AD29	057	PD21/GD5	128	PD40
002	AD28	058	PD26	129	PD38
003	AD27	059	PD20/GD4	130	PD41
004	AD26	061	PD27	131	PD37
005	AD25	063	PD19/GD3	132	PD42
006	AD24	064	PD28	133	PD36
007	C/BE3	065	PD18/GD2	134	PD43
008	IDSEL	066	PD29	136	PD35
010	AD23	067	PD17/GD1	138	PD44
011	AD22	068	PD30	139	PD34
013	AD21	069	PD16/GD0	140	PD45
014	AD20	070	PD31	141	PD33
015	AD19	082	CAS3/DQM3	142	PD46
016	AD18	083	CAS2/DQM2	143	PD32
017	AD17	084	RAS0/SDCLK	144	PD47
018	AD16	085	CAS1/DQM1	146	LD0/PA0
019	C/BE2	086	CAS0/DQM0	147	LD1/PA1
020	FRAME	088	PD7/GA7	148	LCLK/VCLK
021	IRDY	089	PD8/GA8	148	*LCLK/VCLK
023	TRDY	090	PD6/GA6	151	ENFEAT/GOP0
024	DEVSEL	091	PD9/GA9	153	ROMEN//ROMEN/SPD
025	STOP	092	PD5/GA5	154	LD2/PA2
026	PAR	093	PD10/GA10	155	LD3/PA3
027	C/BE1	094	PD4/GA4	174	LD4/PA4
028	AD15	095	PD11/GA11	175	LD5/PA5
029	AD14	097	PD3/GA3	177	N/C//CS
030	AD13	098	PD12/GA12	179	N/C//DSF
031	AD12	100	PD2/GA2	183	ESYNC//N/C
033	AD11	101	PD13/GA13	184	LD6/PA6
034	AD10	102	PD1/GA1	185	LD8/PA8
035	AD9	103	PD14/GA14	186	LD9/PA9
036	AD8	104	PD0/GA0	187	LD10/PA10
038	C/BE0	105	PD15/GA15	188	LD11/PA11
039	AD7	106	PD55	189	LD12/PA12
040	AD6	107	PD56	190	STWR/GOP1
041	AD5	108	PD54	191	BLANK
042	AD4	109	PD57	194	SCLK
043	AD3	110	PD53	196	VCLKI
044	AD2	111	PD58	197	Reserved (PCI GNT)
045	AD1	112	PD52	198	Reserved (PCI REQ)
046	AD0	113	PD59	199	LD13/PA13
048	CAS4/DQM4	115	PD51	200	LD14/PA14
049	CAS5/DQM5	116	PD60	201	LD15/PA15
050	RAS1/OE1/SDRAS	117	PD50	202	LD7/PA7
051	CAS6/DQM6	118	PD61	203	VREQ/VRDY/HREF/
052	CAS7/DQM7	119	PD49		EVIDEO/ESYNC
053	PD23/GD7	120	PD62	204	CREQ/CRDY/VS/EVCLK

Pin names are given as DX name//GX name for the cases where the names are different.
* Pin #148 is connected to two consecutive NAND gates in the NAND tree.

Section 4: Register Changes from the Trio64V+

4.1 REGISTERS CHANGED

Register	Bit(s)	Description
SR1	4	When this bit is set, up to 3 HSYNCs may elapse before the screen is turned off. Software must be careful to allow this amount of time to pass before writing to registers that must be programmed during screen off.
SRA	4-0	LPB N parameter. These are the 5 low order bits of a six bit value controlling the time the LPB has control of the memory bus. The most significant bit of this field is always 1.
SRD	1	0 = No function (Trio64-compatible VAFC feature connector eliminated)
SR12	7	Third bit of the PLL R divider. When SR12_7-5 = 100b, the frequency divider is 16.
SR1C	1-0	The VL-Bus selections are eliminated.
CR2D	7-0	Device ID High = 89H
CR2E	7-0	Device ID Low = 01H
CR2F	7-0	Revision ID = xxH (varies with stepping)
CR32	3	Reserved = 0. This bit should never be set to 1.
CR36	1-0	Reserved - Hardwired to boot in PCI mode
CR36	3-2	00 = 1-cycle EDO 01 = SGRAM 10 = 2-cycle EDO 11 = Fast page
CR36	4	Reserved (no VL-Bus)
CR37	0	Reserved (no VL-Bus)
CR37	1	NAND Tree Testing 0 = Enable NAND tree testing 1 = Normal operation
CR37	2	Subsystem ID Source 0 = Read subsystem ID information from chip registers 1 = Read subsystem ID information from BIOS
CR37	4	Reserved (no VL-Bus)
CR40	0	This bit must only be programmed during screen off (SR1_5 = 1) or during the vertical retrace period.Setting SR1_5 to 1 may take up to 3 HSYNCs to take effect.

CR40	4	Reserved (no VL-Bus)
CR52	2	No VL-Bus General Input Port
CR53	4-3	No VL-Bus power-on default
CR54	2	This is the lsb of the M parameter (not the msb as before)
CR58	3	Reserved (no VL-Bus)
CR66	0	This bit must only be programmed during screen off (SR1_5 = 1) or during the vertical retrace period.Setting SR1_5 to 1 may take up to 3 HSYNCs to take effect.
CR68	2	SGRAM tRP (Refresh to Actiive Command Period) 0 = 4t 1 = 3t
CR68	3	SGRAM tRC (Precharge Command Period) 0 = 10t 1 = 9t
CR68	7	Reserved
CR69	3	Reserved (2 MByte maximum memory)
CR6A	5	Reserved (2 MByte maximum memory)
CR6F	0	0 = Trio64V2/DX mode and pinout 1 = Trio64V2/GX mode and pinout
CR6F	3	PD Hold Time (SGRAM only) 0 = Adjust PD hold time 1 = No effect The BIOS must program this to 0 at power-up for SGRAM configurations.
BEE8H_D		The power-on default value is changed to “Undefined”.
BEE8H_E		The power-on default value is changed to “Undefined”. Software must initialize this register appropriately before the Graphics Engine is used. This is a two step process as explained in the description for BEE8H, read only.
MM8184	29	Color Keying on Index 0 = Color keying based on color value 1 = Color keying based on comparison of color index value with value specified in MM8184_7-0
MM8190	31	Luma-only Interpolation 0 = All YUV values interpolated when vertical filtering enabled 1 = Only Y (luma) values interpolated when vertical filtering enabled
MM81CC	7	Enable Register Load 0 = No effect 1 = Enable function of CR66_5 = 1 (load registers on VSYNC) This bit is automatically cleared to 0 after being set.
MM81E8	13	Memory Bandwidth Saving Mode 1 0 = Disable memory bandwidth savings mode 1 1 = Enable memory bandwidth savings mode 1 This bit is effective when vertical filtering is disabled. This function should not be used if the line length is greater than 720 bytes.

MM81E8	14	Memory Bandwidth Savings Mode 2 0 = Disable memory bandwidth savings mode 2 1 = Enable memory bandwidth savings mode 2 This bit is effective when vertical filtering is enabled.
MM81E8	15	Vertical Filtering 0 = Line replication used 1 = Vertical filtering (interpolation) enabled
MM81E8	18	Reserved
MM81EC	4-0	Streams FIFO Sizes 00000 = Primary Stream = 64 slots; Secondary Stream_NEXT = 0 slots 00100 = Primary Stream = 48 slots; Secondary Stream_NEXT = 16 slots 01000 = Primary Stream = 40 slots; Secondary Stream_NEXT = 24 slots 01100 = Primary Stream = 32 slots; Secondary Stream_NEXT = 32 slots 10000 = Primary Stream = 24 slots; Secondary Stream_NEXT = 40 slots 10100 = Primary Stream = 20 slots; Secondary Stream_NEXT = 44 slots 11000 = Primary Stream = 16 slots; Secondary Stream_NEXT = 48 slots 11100 = Primary Stream = 0 slots; Secondary Stream_NEXT = 64 slots (the following apply only when vertical filtering is disabled) 00010 = Primary Stream = 64 slots; Secondary Stream = 90 slots 11110 = Primary Stream = 90 slots (Enhanced mode only) Each slot holds one quadword.
MM81EC	9-5	Value = (# of secondary stream FIFO slots)/F, where F = 2 if the secondary stream is using the 64-slot FIFO and 4 if it is using the 90-slot FIFO. In other words, each unit represents 2 or 4 FIFO slots
MM81EC	14-10	Value = (# of primary stream FIFO slots)/F, where F = 2 if the primary stream is using the 64-slot FIFO and 4 if it is using the 90-slot FIFO. In other words, each unit represents 2 or 4 FIFO slots
MMFF00	3-1	011 = Reserved (no Video 8 in/out mode)
MMFF00	7	Horizontal Decimation Type 0 = Trio64V+-type horizontal decimation 1 = New horizontal decimation type described in Section 3.2.1.
MMFF00	14	Auto Sync Generation 0 = No auto sync generation 1 = HSYNC and VSYNC automatically generated based on the horizontal count programmed in MMFF28_11-0 and the vertical count programmed in MMFF28_24-16. When the horizontal count value is reached, the stride (MMFF34_11-0) is added to the address to start the next line. This bit is used for LPB pass-through mode.

MMFF00	20	Video HSYNC 0 = Horizontal window size value used to reset horizontal counter and start a new line in Video 8 and 16 modes 1 = HREF input used to reset horizontal counter and start a new line in Video 8 and 16 modes. This register must be programmed to 1 when the width of an incoming live video line may be less than the programmed size.
MMFF00	23-21	Video FIFO Threshold 000 = 1 slot 001 = 2 slots 010 = 4 slots 011 = 6 slots 100 = 8 slots 101 = 10 slots 110 = 12 slots 111 = 14 slots When the number of video FIFO slots programmed in this register are filled, a request is made to the memory controller to empty the FIFO.
MMFF00	28	Odd/even field Indicator (Read Only) 0 = Odd field being processed 1 = Even field being processed This bit is valid when interlaced video data is being input via the LPB using Video 8 or Video 16 mode.
MMFF00	29	Odd/Even Field Indicator Invert 0 = Odd/even field definition specified by bit 28 of this register 1 = Odd/even field definition inverted from the definition in bit 28 of this register
MMFF00	30	Odd/Even Field Detection Enable 0 = Odd/even field detection disabled 1 = Odd/even field detection enabled This function can be enabled when the LPB is operating in its Video 8 or Video 16 mode.
MMFF00	31	Reserved (no CL-480 support)
MMFF04	4	Video FIFO Flush 0 = No Video FIFO flush 1 = Video FIFO flush in process This bit is used for test purposes only.
MMFF04	7-5	LPB FIFO Write Pointer Status Value = State of LPB FIFO write pointer This bit is used for test purposes only.
MMFF04	10-8	LPB FIFO Read Pointer Status Value = State of LPB FIFO read pointer This bit is used for test purposes only.

MMFF04	18-14	Video FIFO Write Pointer Status Value = State of video FIFO write pointer This bit is used for test purposes only.
MMFF04	27-23	video FIFO Read Pointer Status Value = State of Video FIFO read pointer This bit is used for test purposes only.
MMFF04	28	Frame Dropping Status 0 = Current frame dropped 1 = Current frame being passed to memory
MMFF28	11-0	Auto Sync Horizontal Size Value = (number of pixels/line)/2 This value is valid only for pass-through mode and is used to generate automatic HSYNCs. The original definition for these bits still applies for Video 8 or 16 mode.
MMFF28	24-16	Auto Sync Vertical Size Value = number of lines of video per frame This value is valid only for pass-through mode and is used to generate automatic VSYNCs. The original definition for these bits still applies for Video 8 or 16 mode.
MMFF2C	31-0	Decimation for 4:2:2 input only. See the description in Section 3.
PCI Index 02H	15-0	Device ID = 8901H
PCI Index 08H	7-0	Revision ID = xxH (varies with stepping)

4.2 NEW REGISTERS

Extended Sequencer 1A Register (SR1A)

Read/Write Address: 3C5H, Index 1AH
Power-On Default: 00H

7	6	5	4	3	2	1	0
R	R	BP	FCTD	DCLK PLLA		MCLK PLLA	

Bits 1-0 MCLK PLLA - MCLK PLL Adjust

Value = PLL gain adjustment factor

This function is used for S3 testing.

Bits 3-2 DCLK PLLA - DCLK PLL Adjust

Value = PLL gain adjustment factor

This function is used for S3 testing.

Bit 4 FCTD - Filter Capacitor Test Disable
0 = No effect
1 = Filter capacitor test mode disabled

This function is used for S3 testing.

Note: This bit must be set to 1 by the BIOS upon reset.

Bit 5 BP - BLANK Pedestal
0 = No $\overline{\text{BLANK}}$ pedestal
1 = Add $\overline{\text{BLANK}}$ Pedestal to RAMDAC output

Bits 7-6 Reserved

Extended Sequencer 1B Register (SR1B)

Read/Write Address: 3C5H, Index 1BH
Power-On Default: 00H

7	6	5	4	3	2	1	0
R	R	R	R	R	DAC OUTPUT		

Bits 2-0 DAC OUTPUT - RAMDAC Output Current Control

The higher the value in this field, the lower the maximum DAC output current.

This function is used for S3 testing.

Bits 7-3 Reserved

Synchronization 1 Register (CTR23)

Read/Write Address: 3?5H, Index 23H
Power-On Default: 00H

This register must be 00H before CR26 is written. For this to be effective, A5H must first be programmed into CR39.

7	6	5	4	3	2	1	0
R	R	R	R	R	R	R	R

Bits 7-0 Reserved

Synchronization 2 Register (CTR26)

Read Only Address: 3?5H, Index 26H
Power-On Default: 00H

The BIOS must write 00H to this register upon each mode set (assuming CR23 is at its default value of 00H). Drivers should write 00H to this register after writing 00H to CR23 before enabling vertical interpolation. For this to be effective, A5H must first be programmed into CR39.

7	6	5	4	3	2	1	0
R	R	R	R	R	R	R	R

Bits 7-0 Reserved

Extended Memory Control 5 Register (CR72)

Read/Write Address: 3?5H, Index 72H
Power-On Default: 00H

7	6	5	4	3	2	1	0
SECONDARY STREAM FIFO N PARAMETER							

Bits 7-0 SECONDARY STREAM FIFO N PARAMETER

Value = Number of MCLKs allocated to screen refresh FIFO filling before control of the memory bus is relinquished.

This value is effective only when the FIFO is at or above its threshold value (low priority).

Drive Current Control Register (CR80)

Read/Write Address: 3?5H, Index 80H
Power-On Default: 00H

7	6	5	4	3	2	1	0
R	R	R	PDD	R	R	R	R

Bits 3-0 Reserved

Bit 4 PDD - PD Drive Current
0 = PD[63:0] drive current is 8 ma
1 = PD[63:0] drive current is 4 ma

Bits 7-5 Reserved

PCI Subsystem Vendor ID Shadow Low Register (CR81)

Read/Write Address: 3?5H, Index 81H
Power-On Default: 00H

7	6	5	4	3	2	1	0
PCI SUBSYSTEM VENDOR ID LOW BYTE							

Bits 7-0 PCI SUBSYSTEM VENDOR ID LOW BYTE

This register shadows the byte at PCI configuration space index 2CH.

PCI Subsystem Vendor ID Shadow High Register (CR82)

Read/Write Address: 3?5H, Index 82H
Power-On Default: 00H

7	6	5	4	3	2	1	0
PCI SUBSYSTEM VENDOR ID HIGH BYTE							

Bits 7-0 PCI SUBSYSTEM VENDOR ID HIGH BYTE

This register shadows the byte at PCI configuration space index 2DH.

PCI Subsystem ID Shadow Low Register (CR83)

Read/Write Address: 3?5H, Index 83H
Power-On Default: 00H

7	6	5	4	3	2	1	0
PCI SUBSYSTEM ID LOW BYTE							

Bits 7-0 PCI SUBSYSTEM ID LOW BYTE

This register shadows the byte at PCI configuration space index 2EH.

PCI Subsystem ID Shadow High Register (CR84)

Read/Write Address: 3?5H, Index 84H
Power-On Default: 00H

7	6	5	4	3	2	1	0
PCI SUBSYSTEM ID HIGH BYTE							

Bits 7-0 PCI SUBSYSTEM ID HIGH BYTE

This register shadows the byte at PCI configuration space index 2FH.

FIFO Fetch Delay Register (CR85)

Read/Write Address: 3?5H, Index 85H
Power-On Default: 00H

7	6	5	4	3	2	1	0
R	R	R	R	R	FIFO FETCH DELAY		

Bits 2-0 FIFO FETCH DELAY

Value = number of character clocks to delay start of FIFO fetching.

This value only applies to full Streams Processor mode (CR67_3-2 = 11b), where FIFO fetching automatically begins as soon as feasible after the start of blanking.

Bits 7-3 Reserved

DAC Power Up Register (CR86)

Read/Write Address: 3?5H, Index 86H
Power-On Default: 00H

7	6	5	4	3	2	1	0
DPSD	DAC POWER UP TIME						

Bits 6-0 DAC POWER UP TIME

value = number of character clocks from the start of blanking at which the internal DACs are powered up

This value must be at least 1 less than the End Horizontal Blank value programmed in CR5D_3, CR5_7 and CR3_4-0. A value of 1 less starts DAC power up 1 character clock before the end of blanking. A value of 2 less starts DAC power up 2 character clocks before the end of blanking, etc. When the DACs power up, there is a voltage spike that affects the RGB outputs if they are active. Powering up the DACs earlier reduces the power savings but also reduces the chance that the power up voltage spike will affect the active display.

Bit 7 DPSD - DAC Power Saving Disable
0 = RAMDAC power saving enabled (DAC turned off at BLANK start and on at position programmed in bits 6-0 of this register
1 = RAMDAC power saving disabled (RAMDAC never powered down)

Miscellaneous Control Register (CR87)

Read/Write Address: 3?5H, Index 87H
Power-On Default: 00H

7	6	5	4	3	2	1	0
R	R	R	R	R	R	1LA	DUV

Bit 0 DUV - Drop UV/CbCr
0 = Pass UV/CbCr color components to the line buffer
1 = Drop UV/CbCr color components before writing to the line buffer

This bit is used to reduce the bandwidth requirements for live video. It is effective when vertical filtering is enabled (MM81E8_15 = 1) and memory bandwidth savings mode is enabled (MM81E8_14 = 1).

Bit 1 1LA - 1-cycle Linear Addressing
0 = Disable 1-cycle linear addressing
1 = Enable 1-cycle linear addressing write operation

If this bit is set, the video memory must be capable of supporting 1-cycle operation.

Bits 7-2 Reserved

Synchronous DRAM Control Register (CR88)

Read/Write Address: 3?5H, Index 88H
Power-On Default: 00H

This register is valid only for the Trio64V2/GX.

7	6	5	4	3	2	1	0
R	DR	SMS	SCL		R	PDD	R

Bit 0 Reserved

Bit 1 PDD - PD Bus Delay
0 = No delay
1 = Delay PD bus output by 1/2 MCLK (memory writes)

Bit 2 Reserved

Bits 4-3 SCL - SGRAM CAS Latency
00 = Reserved
01 = Reserved (for 1 MCLK)
10 = 3 MCLKs
11 = 2 MCLKs

Bit 5 SMS - SGRAM Set Mode Register
After the $\overline{CAS}$ latency is programmed, this bit is programmed to 1 to generate a mode programming cycle. This bit is automatically cleared to 0 after the programming cycle.

Bit 6 DR - Disable RAM Refresh
0 = RAM refresh enabled
1 = RAM refresh disabled

Bit 7 Reserved

Primary Stream FIFO Fetch Control 1 Register (CR90)

Read/Write Address: 3?5H, Index 90H
Power-On Default: 00H

7	6	5	4	3	2	1	0
EL1	R	R	R	R	L1 10-8		

Bits 2-0 L1 10-8 - Primary Stream L1 Parameter (Bits 10-8)
These are bits 10-8 of the primary stream L1 parameter. See the description of the primary stream L1 parameter in the description for CR91.

Bits 6-3 Reserved

Bit 7 EL1 - Enable L1 Parameter
0 = Primary stream display fetch length control (L1 parameter) disabled
1 = Primary stream display fetch length control (L1 parameter) enabled

Primary Stream FIFO Fetch Control 2 (CR91)

Read/Write Address: 3?5H, Index 91H
Power-On Default: 00H

These are the lower 8 bits of an 11-bit value used in conjunction with the M and N parameters to optimize performance. If the primary stream display FIFO is being filled when this value is reached, FIFO filling will be stopped and memory control will be switched to the M parameter. The upper three bits are bits 2-0 of CR90.

7	6	5	4	3	2	1	0
L1 7-0							

Bits 7-0 L1 7-0 - Primary Stream L1 Parameter (Bits 7-0)
11-bit Value = [(number of bytes of displayed pixels per scan line) ÷ 8] - 1. This register contains the least significant 8 bits of this value.

Secondary Stream FIFO Fetch Control 1 Register (CR92)

Read/Write Address: 3?5H, Index 92H
Power-On Default: 00H

7	6	5	4	3	2	1	0
EL2	R	R	R	R	L2 10-8		

Bits 2-0 L2 10-8 - Secondary Stream L2 Parameter (Bits 10-8)
These are bits 10-8 of the secondary stream L2 parameter. See the description of the primary stream L2 parameter in the description for CR93.

Bits 6-3 Reserved

Bit 7 EL2 - Enable L2 Parameter
0 = Secondary stream display fetch length control (L2 parameter) disabled
1 = Secondary stream display fetch length control (L2 parameter) enabled

Secondary Stream FIFO Fetch Control 2 (CR93)

Read/Write Address: 3?5H, Index 93H
Power-On Default: 00H

These are the lower 8 bits of an 11-bit value used in conjunction with the M and N parameters to optimize performance. If the secondary stream display FIFO is being filled when this value is reached, FIFO filling will be stopped and memory control will be switched to the M parameter. The upper three bits are bits 2-0 of CR92.

7	6	5	4	3	2	1	0
L2 7-0							

Bits 7-0 L2 7-0 - Secondary Stream L2 Parameter (Bits 7-0)
11-bit Value = [(number of bytes of displayed pixels per scan line) ÷ 8] - 1. This register contains the least significant 8 bits of this value.

Color Adjustment Register (MM819C)

Read/Write Offset: 819C
Power-on Default: 00000000H

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BCE	R	R	CONTRAST					BRIGHTNESS							
31	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**
HSE	R	R	HUE/SAT 2					R	R	R	HUE/SAT 1				

Bits 7-0 BRIGHTNESS - Brightness Control

Value = SBBBBBBB

where S is the sign bit (1 = negative) and BBBBBBB is the brightness adjustment factor. The larger the number, the greater the brightness. The value can range from -256 to 254 because the programmed value is shifted left by 1, thereby multiplying the value by 2. The value is in 2's complement format.

Bits 12-8 CONTRAST- Contrast Control

Value = C.CCCC

This is the contrast adjustment, which can vary from 0 to 1.9375. The value is in 2's complement format.

Bits 14-13 Reserved

Bits 15 BCE - Brightness and Contrast Enable
0 = Brightness and contrast control disabled
1 = Brightness and contrast control enabled

This control should be enabled only for YUV/YCbCr secondary stream formats and must be disabled for RGB secondary stream formats.

Bits 20-16 HUE/SAT 1 - Hue and Saturation Factor 1

Value = SF.FFF

where S is the sign bit (1 = negative) and F.FFF is the factor [SAT * cosine A]. SAT is the saturation, which can vary from -2 to 1.85 and A is the hue angle, the cosine of which can vary from -1 to +1. The value is in 2's complement format.

Bits 23-21 Reserved

Bits 28-24 HUE/SAT 2 - Hue and Saturation Factor 2

Value = SF.FFF

where S is the sign bit (1 = negative) and F.FFF is the factor [SAT * sine A]. SAT is the saturation, which can vary from -2 to 1.85 and A is the hue angle, the sine of which can vary from -1 to +1. The value is in 2's complement format.

Bits 30-29 Reserved

Bit 31 HSE - Hue and Saturation Control Enable
0 = Hue and saturation control disabled
1 = Hue and saturation control enabled

This control should be enabled only for YUV/YCbCr secondary stream formats and must be disabled for RGB secondary stream formats.

PCI Configuration Space Subsystem ID

Read Only Address: 2CH
Power-On Default: 00000000H

This register is a shadow of CR81-CR84.

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SUBSYSTEM VENDOR ID															
31	**30**	**29**	**28**	**27**	**26**	**25**	**24**	**23**	**22**	**21**	**20**	**19**	**18**	**17**	**16**
SUBSYSTEM ID															

Bits 15-0 SUBSYSTEM VENDOR ID

Bits 31-16 SUBSYSTEM ID

Section 5: Electrical Data

5.1 MAXIMUM RATINGS

Table 5-1. Absolute Maximum Ratings

Ambient temperature	0° C to 70° C
Storage temperature	-40° C to 125° C
DC Supply Voltage	-0.5V to 7.0V
I/O Pin Voltage with respect to V_{SS}	-0.5V to V_{DD}+0.5V

5.2 DC SPECIFICATIONS

Note: In all cases below, digital VDD = 5V ± 5% and the operating temperature is 0° C to 70° C.

Table 5-2. RAMDAC/Clock Synthesizer DC Specifications

Symbol	Parameter	Min	Typical	Max	Unit
AVDD	DAC supply voltage	4.75	5	5.25	V
AVDD (CLOCK)	PLL supply voltage	4.75	5	5.25	V
VREF	Internal voltage reference	1.10	1.235	1.35	V

Table 5-3. RAMDAC Characteristics

	Min	Typical	Max	Unit
Resolution Each DAC		8		bits
LSB Size		66		µA
Integral Linearity Error			± 1	LSB
Differential Linearity Error			± 1	LSB
Output Full-Scale Current	15.4	17.6	19.8	mA
DAC to DAC Mismatch			5%	
Power Supply Rejection Ratio			0.5	%/ % AVDD
Output Compliance	0.0		1.5	V
Output Capacitance			30	pF
Glitch Impulse		75		pV-Sec

Table 5-4. Digital DC Specifications (VDD = 5V ± 5%, Operating Temperature 0° C to 70° C)

Symbol	Parameter	Min	Max	Unit
V_{IL}	Input Low Voltage	-0.5	0.8	V
V_{IH}	Input High Voltage	2.4	V_{DD} +0.5	V
V_{OL}	Output Low Voltage		V_{SS} + 0.4	V
V_{OH}	Output High Voltage	2.4		V
I_{OL1}	Output Low Current	8 (Note 1)		mA
I_{OH1}	Output High Current	-4		mA
I_{OL2}	Output Low Current	16 (Note 2)		mA
I_{OH2}	Output High Current	-8		mA
I_{OL3}	Output Low Current	24 (Note 3)		mA
I_{OH3}	Output High Current	-12		mA
I_{OZ}	Output Tri-state Current		1	µA
C_{IN}	Input Capacitance		5	pF
C_{OUT}	Output Capacitance		5	pF
I_{CC}	Power Supply Current		TBD (Note 4)	mA

Notes

1. I_{OL1}, I_{OH1} for pins $\overline{\text{ROMEN}}$, $\overline{\text{INTA}}$, $\overline{\text{STWR}}$, HSYNC, VSYNC, VCLK, $\overline{\text{BLANK}}$, $\overline{\text{ENFEAT}}$, MA[8:0], $\overline{\text{CAS}}$[7:0], PD[63:0], AD[31:0], LD[7:0], $\overline{\text{VREQ}}$/VRDY, SPCLK, SPD
2. I_{OL2}, I_{OH2} for pins $\overline{\text{OE}}$[1:0], $\overline{\text{WE}}$, $\overline{\text{RAS}}$[1:0]
3. I_{OL3}, I_{OH3} for pins PAR, $\overline{\text{STOP}}$, $\overline{\text{DEVSEL}}$, $\overline{\text{TRDY}}$
4. I_{CC} measured for a resolution of TBD.

5.3 AC SPECIFICATIONS

5.3.1 RAMDAC AC Specifications

Table 5-5. RAMDAC AC Specifications

Parameter	Typical	Max	Unit	Notes
DAC Output Delay	5		ns	1
DAC Output Rise/Fall Time	3		ns	2
DAC Output Settling Time	15		ns	
DAC-to-DAC Output Skew	2	5	ns	3

Notes

1. Measured from the 50% point of VCLK to the 50% point of full scale transition
2. Measured from 10% to 90% full scale
3 With DAC outputs equally loaded

Table 5-6. RAMDAC Output Specifications

Description	I_{OUT} (mA)	V_{OUT} (V)	BLANK	Input Data
White (with $\overline{\text{BLANK}}$ pedestal)	19.05 (typical)	0.714 (typical)	1	FFH
White	17.6 (typical)	0.66 (typical	1	FFH
Data (with $\overline{\text{BLANK}}$ pedestal)	Data + 1.45	Data + 0.054	1	Data
Data	Data	Data	1	Data
Black (with $\overline{\text{BLANK}}$ pedestal)	1.45 (typical)	0.054	1	00H
Black (no pedestal)	0	0	1	00H
$\overline{\text{BLANK}}$	0	0	0	Don't Care

Notes for Table 5-6

1. Condition for V_{OUT} is a 75 Ohm doubly terminated load, use of the internal VREF and RSET = 147 Ohms.
2, No sync pedestal is provided. Sync output levels are the same as for black output.
3. The $\overline{\text{BLANK}}$ pedestal is active when SR1A_5 = 1.

5.3.2 Clock Timing

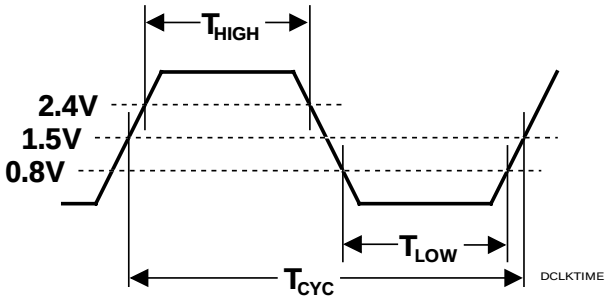


Figure 5-1. Clock Waveform Timing

Table 5-7. Clock Waveform Timing

Symbol	Parameter	Min	Max	Units	Notes
T_{CYC}	SCLK Cycle Time	30	125	ns	1
	LCLK Cycle Time	30	200	ns	
	MCLK Cycle Time (fast page)	15	100	ns	
	MCLK Cycle Time (1-cycle EDO)	16.67	100	ns	
	DCLK Cycle Time (VGA Mode)	25	100	ns	1
	DCLK Cycle Time (Enhanced Mode)	12.5	100	ns	1, 2
	SDCLK Cycle Time	15	100	ns	
T_{HIGH}	SCLK High Time	12	80	ns	
	LCLK High Time	12	160	ns	
	SDCLK High Time	6	80	ns	
T_{LOW}	SCLK Low Time	12	80	ns	
	LCLK Low Time	12	160	ns	
	SDCLK Low Time	6	80	ns	
	SCLK, LCLK, SDCLK Slew Rate	1	4	V/ns	3

Notes

1. $f_{DCLK} \geq 1/2\ f_{SCLK}$ to ensure valid writes to the PLLs.
2. The maximimum DCLK with the Streams Processor enabled and single clocking is 80 MHz. With the Streams Processor disabled and single clocking, the maximum DCLK is 95 MHz. The maximum DCLK rate with clock doubling is 170 MHz, with two 8-bit pixels clocked into the RAMDAC each clock at 85 MHz.
3. Rise and fall times are specified in terms of the edge rate measured in V/ns. This slew rate must be met across the minimum peak-to peak portion of the clock waveform.

5.3.3 Input/Output Timing

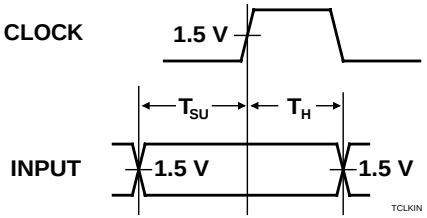


Figure 5-2. Input Timing

Table 5-8. SCLK-Referenced Input Timing

PCI Bus			
Symbol	**Parameter**	**Min**	**Units**
T_{SU}	AD[31:0], $\overline{C/BE[3:0]}$, $\overline{FRAME}$, $\overline{IRDY}$, IDSEL setup	7	ns
T_H	AD[31:0] hold	1	ns
T_H	$\overline{C/BE[3:0]}$, $\overline{FRAME}$, $\overline{IRDY}$, IDSEL hold	1	ns
Miscellaneous			
Symbol	**Parameter**	**Min**	**Units**
T_{SU}	ROM data GD[7:0] setup (PCI)	5	ns
T_H	ROM data GD[7:0] hold (PCI)	7	ns
T_{SU}	General Input Port GD[7:0] setup	5	ns
T_H	General Input Port GD[7:0] hold	7	ns

Table 5-9. LCLK-Referenced Input Timing

Scenic/MX2 Interface			
Symbol	**Parameter**	**Min**	**Units**
T_{SU}	LD[7:0] setup	10	ns
T_H	LD[7:0] hold	9	ns
T_{SU}	$\overline{CREQ}$/CRDY	6	ns
T_H	$\overline{CREQ}$/CRDY	8	ns
SAA7110 Interface			
Symbol	**Parameter**	**Min**	**Units**
T_{SU}	LD[7:0] setup (also LD[15:8] for 16-bit interface)	6	ns
T_H	LD[7:0] hold (also LD[15:8] for 16-bit interface)	8	ns
T_{SU}	HS setup	6	ns
T_H	HS hold	7	ns
T_{SU}	VS setup	6	ns
T_H	VS hold	7	ns

Table 5-10. MCLK-Referenced Input Timing

Symbol	**Parameter**	**Min**	**Units**
T_{SU}	PD[63:0] setup to MCLK high (2-cycle EDO)	0	ns
T_H	PD[63:0] hold from MCLK high (2-cycle EDO)	12.5	ns
T_{SU}	PD[63:0] setup to following $\overline{CAS}$ low (1-cycle EDO)	0	ns
T_H	PD[63:0] hold from following $\overline{CAS}$ low (1-cycle EDO)	15	ns
T_{SU}	PD[63:0] setup to $\overline{CAS}$ high (fast page)	0	ns
T_H	PD[63:0] hold from $\overline{CAS}$ high (fast page)	15	ns

Note

1. The timing reference in each of the three cases above is to the event that causes the latching of the read data. The MCLK used to latch 2-cycle EDO data is an internal signal that cannot be directly observed. The $\overline{CAS}$ signals used to latch read data in the other operational modes are derived from the internal MCLK.

Table 5-11. SDCLK-Referenced Input Timing

Symbol	**Parameter**	**Min**	**Units**
T_{SU}	PD[63:0] setup	13	ns
T_H	PD[63:0] hold	4	ns

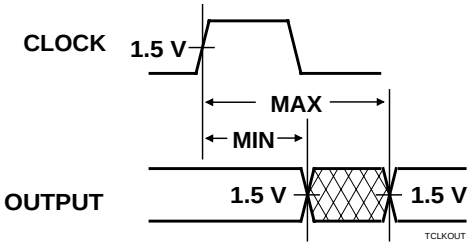


Figure 5-3. Output Timing

The minimum delay is the minimum time after the clock edge that the valid signal state from the previous cycle will begin transition to the next state (become invalid).

The maximum delay is the maximum time after the clock edge that the signal state is valid for the next cycle.

Table 5-12. SCLK-Referenced Output Timing

PCI Bus				
Parameter	**Min**	**Max**	**Units**	**Notes**
AD[31:0] valid delay	2	16	ns	1
$\overline{\text{DEVSEL}}$, PAR delay	2	11	ns	Medium $\overline{\text{DEVSEL}}$ timing used
$\overline{\text{STOP}}$ delay	2	11	ns	
$\overline{\text{TRDY}}$ delay	2	11	ns	
INTA delay	2	11	ns	
Miscellaneous				
Parameter	**Min**	**Max**	**Units**	**Notes**
$\overline{\text{ROMEN}}$ delay	4	10	ns	
ROM address valid delay	5	30	ns	
AD[7:0] ROM data valid delay	5	30	ns	

Note

1. Due to the timing for $\overline{\text{TRDY}}$ for read cycles, data is not sampled on the clock edge immediately following its becoming valid. This guarantees the PCI 2.1 specification time of 11 ns.

Table 5-13. LCLK-Referenced Output Timing

Scenic/MX2 Interface				
Parameter	**Min**	**Max**	**Units**	**Notes**
$\overline{\text{VREQ}}$/VRDY active delay	2	11	ns	7 ns typ
LD[7:0] valid delay	2	15	ns	8 ns typ
LD[7:0] tri-state from LCLK	7	15	ns	

Table 5-14. MCLK-Referenced Output Timing

Parameter	**Min**	**Max**	**Units**	**Notes**
PD[63:0] valid delay	5	8	ns	1
MA[8:0] valid delay	6	8	ns	
$\overline{\text{CAS}}$[7:0] active delay	5	12	ns	
$\overline{\text{CAS}}$[7:0] inactive delay	5	14	ns	
$\overline{\text{RAS}}$[1:0] active delay	2	6	ns	
$\overline{\text{RAS}}$[1:0] inactive delay	0	7	ns	
$\overline{\text{OE}}$[1:0] active delay	6	12	ns	
$\overline{\text{WE}}$ active delay	7	12	ns	

Note

1. The maximum delay time is for 1-cycle operation.

Table 5-15. SDCLK-Referenced Output Timing

Parameter	**Min**	**Max**	**Units**	**Notes**
PD[63:0] valid delay	2	8	ns	
MA[9:0] valid delay	5	9	ns	
DQM[7:0] active delay	4	8	ns	
DQM[7:0] inactive delay	8	13	ns	
$\overline{\text{SDRAS}}$ active delay	4	8	ns	
$\overline{\text{SDRAS}}$ inactive delay	6	11	ns	
$\overline{\text{SDCAS}}$ active delay	4	8	ns	
$\overline{\text{SDCAS}}$ inactive delay	6	11	ns	
CS active delay	4	8	ns	
$\overline{\text{CS}}$ inactive delay	6	12	ns	
$\overline{\text{WE}}$ active delay	4	8	ns	
$\overline{\text{WE}}$ inactive delay	6	12	ns	

Table 5-16. Feature Connector Timing - Output from Trio64V2 to Feature Connector

Symbol	Parameter	Min	Units	Notes
T_{SU}	PA[15:0], $\overline{BLANK}$ setup to VCLK rising	5	ns	
T_H	PA[15:0], $\overline{BLANK}$ hold from VCLK rising	5	ns	

Table 5-17. Feature Connector Timing - Output from Feature Connector to Trio64V2

Symbol	Parameter	Min	Max	Units	Notes
T_{SU}	PA[15:0], $\overline{BLANK}$ setup to VCLK rising	6		ns	
T_H	PA[15:0], $\overline{BLANK}$ hold from VCLK rising	6		ns	
	VCLK	25	40	ns	1
	VCLK duty cycle	40	60	%	
	VCLK high time	10	25	ns	
	VCLK low time	10	25	ns	
	VCLK slew rate	1	4	V/ns	

Notes

1. This corresponds to the VESA VAFC specification of a maximum clock of 37.5 MHz.

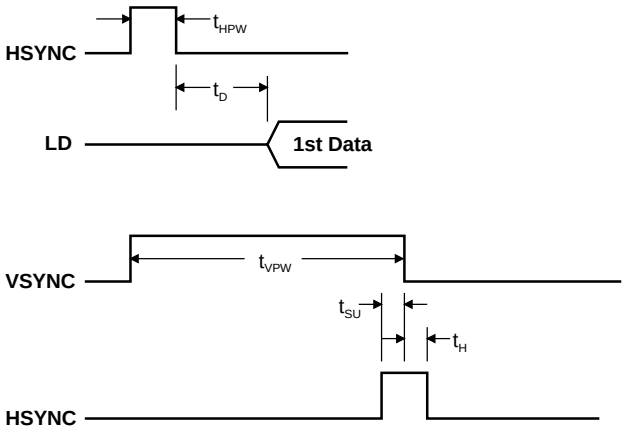


Figure 5-4. Odd/Even Field Detection Timing

Table 5-18. Odd/Even Field Detection Timing

Symbol	Parameter	Min	Max
T_D	Time from HSYNC falling edge to 1st data	2 µs	
T_{SU}	HSYNC high to VSYNC falling edge	3 ns	
T_H	HSYNC high after VSYNC falling edge	3 ns	
T_{HPW}	HSYNC pulse width	1 LCLK	6 µs
T_{VPW}	VSYNC pulse width	1 LCLK	528 µs

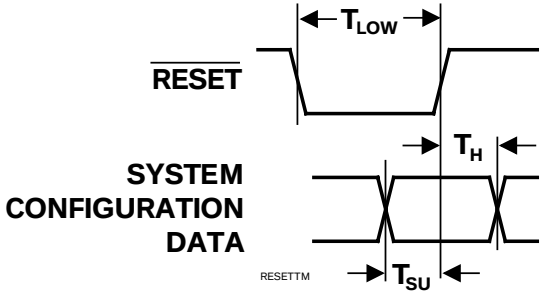


Figure 5-5. Reset Timing

Table 5-19. Reset Timing

Symbol	Parameter	Min	Units
T_{LOW}	$\overline{RESET}$ active pulse width	1000	ns
T_{SU}	PD[28:0] setup to $\overline{RESET}$ inactive	20	ns
T_H	PD[28:0] hold from $\overline{RESET}$ inactive	10	ns

Section 6: Mechanical Data

6.1 THERMAL SPECIFICATIONS

Parameter	Min	Typ	Max	Unit
Thermal Resistance Θ_{JC}		5		°C/W
Thermal Resistance Θ_{JA} (Still Air)		24		°C/W
Junction Temperature			125	°C

6.2 MECHANICAL DIMENSIONS

Trio64V2DX and Trio64V2/GX come in 208-pin PQFP packages. The mechanical dimensions are given in Figure 6-1.

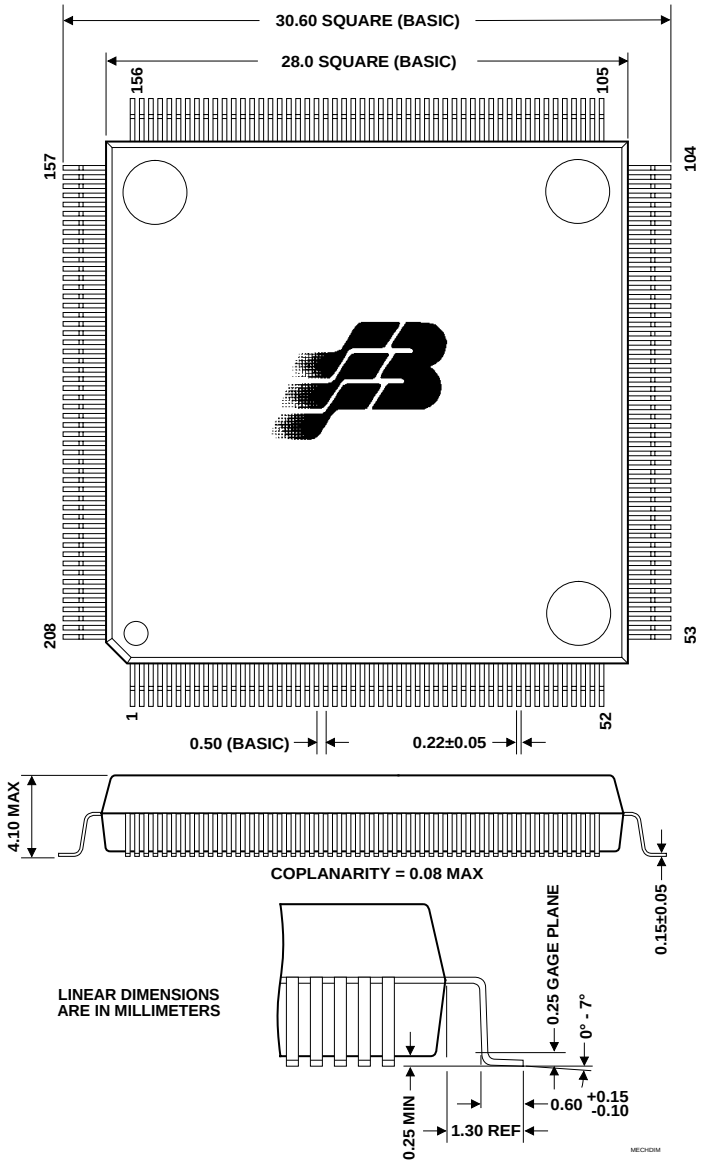


Figure 6-1. 208-pin PQFP Mechanical Dimensions

S3 Incorporated, P.O. Box 58058, Santa Clara, CA 95052-8058 Tel: 408-588-8000, Fax: 408-980-5444